SCIENCE

Astronomy for beginners

Hereward Carrington

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Astronomy for Beginners

Hereward Carrington, Ph.D.

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CONTENTS.

Page

Introductory 5

Astrology 7

The Solar System 8

The Sun 9

Mercury 11

Venus 12

The Earth 13

Mars 13

Jupiter 16

Saturn 17

Uranus 18

Neptune 18

The Minor Planets 19

Are There Other Planets? 20

The Moon 20

The Origin of the Solar System 24

Constellations 27

Meteors: “Shooting Stars” 28

Meteorites 29

Comets 30

Nebulae 32

The Milky Way 33

The Number of Stars 33

The Position of Our Solar System 35

The Movement of Our Solar System 36

Distances of the Stars 36

Temperature of the Stars 37

Fixed Stars 37

Double Stars 38

Colored Stars 38

Variable Stars 39

Temporary Stars 40

Star Groups--Clusters 40

Eclipses 41

Telescopes 43

The Spectroscope: Spectrum Analysis 43

Photography 46

The Tides 47

Gravitation 48

The Ether 49

Atomic Analogies 49

Thunder and Lightning 50

Fireballs 50

Atmospheric Electricity 50

The Earth’s Magnetism 51

The Aurora Borealis 51

Time: Measurement of 52

Space: Measurement of 52

The International Day Line 54

Calendars, etc. 54

Curved Space 55

The Temperature of Space 55

Light in Space 55

Life in Space 56

The Causes of an Ice Age 56

Why Do Stars “Twinkle”? 56

Why Does the Moon Sometimes Appear Larger? 57

Are the Planets Inhabited? 57

A Few Definitions 58

INTRODUCTORY

Astronomy is one of the oldest of the sciences--as it is one of the most fascinating! The early Egyptians, Assyrians, Babylonians and Chaldeans were, as we know, great astronomers, and, considering that they were compelled to make their observations without the aid of telescopes, some of their conclusions are truly remarkable in their accuracy. Men must always have gazed at the stars, and wondered at their number and their beauty; yet it is only within the past three or four centuries that accurate ideas as to the nature, size and structure of our Universe have come into being. No subject is so calculated to impress upon man his own relative insignificance as astronomy--to show him that the speck of matter upon which he dwells is so small that it cannot even be seen, from a relatively short distance in space! How puny and absurdly trifling seem his bickerings and his disputes, his wars and his hates, his jealousies and his failures, when viewed from the standpoint of infinite time and infinite space; mere struggles upon an ant hill, which, a few million years from now, will be uninhabitable, while the sublime immensity of Nature will proceed as if nothing had happened!

Yes, astronomy is a fascinating and romantic study, and the following little book is an attempt to summarize, very briefly, the most important findings of modern astronomical science upon this question. I have endeavored to make the subject as simple as possible, and to avoid all terms of a technical character, unless these are fully explained. It is my hope that the reader may be enabled to gain a fairly clear and accurate idea as to the nature and constitution of our Universe by a perusal of this little book.

ASTRONOMY FOR BEGINNERS

ASTROLOGY

Those who have not studied this subject will often ask the question: “What is the difference between Astronomy and Astrology?” It is merely this: Astronomy studies the heavenly bodies, and their movements, etc., by all available scientific means; while astrology, also utilizing this material, further asserts that the particular relative positions which the sun, moon, planets and other heavenly bodies occupy at the

moment of birth _influence the individual_ born at that moment, and continue to influence him all through life. In other words, astrology is undoubtedly an exact science in so far as its astronomical _data_ are concerned; but its further inference, as applied to the living human being, is not; it depends upon historic beliefs and traditions which have been handed-down for centuries. So far as astronomers have been enabled to ascertain, there is not the slightest scientific basis for any belief in astrology; assuredly it is a curious and interesting occult study, but it must be understood to lie within that realm, rather than in that of exact astronomical science.

THE SOLAR SYSTEM

When we speak of the “solar system,” we mean our central Sun, and the various planets which revolve around it. The planets, beginning with the one nearest the Sun, and proceeding outwards into space, are: Mercury, Venus, The Earth, Mars, Jupiter, Saturn, Uranus, Neptune. Most of these planets have in turn, circling round them, smaller bodies of satellites; the Earth has but one--the Moon. Other planets have more than one. The planets vary greatly in size, as well as in their relative distances from the Sun. The following may help the reader to form a mental picture of their relative distances and sizes:

Imagine a large open common; on it place a globe 2 feet in diameter, by way of representing the Sun; Mercury will then be represented by a mustard seed at a distance of 82 feet; Venus by a pea at a distance of 142 feet; the Earth also by a pea, at a distance of 215 feet; Mars will be a small pepper corn, at a distance of 327 feet; the Minor Planets by grains of sand at distances varying from 500 to 600 feet; then a moderate sized orange ¼ of a mile distant from the central point will represent Jupiter; a small orange ⅖ths of a mile, Saturn; a full-sized cherry, ¾ths of a mile, Uranus; and lastly a plum, at 1¼ miles, Neptune,--the most distant planet yet known,--though some astronomers suspect there may exist another planet still further off, and hope one day to find it. (On the same scale, the _nearest_ “fixed” star would be 7,500 miles distant).

THE SUN

The Sun is the center of our solar (sun) system; it is the great giver of light and heat, without which life upon our planet would soon become extinct. It is an immense body, more than a million times the size of our earth. In fact, its _radius_ is nearly twice the distance of the moon from the earth! The mass of the Sun is 332,000 times that of the earth. It gives 600,000 times as much light as the full moon. The energy radiated per square yard from the Sun is equivalent to 140,000 horse power. The heat radiated by the sun would melt a layer of ice 4,000 feet thick every hour, all over its surface. Various estimates of the amount of heat upon the surface have been made, but these do not agree,--figures all the way from 10,000° F. to 180,000° F. having been given. Certain it is that its internal heat is terrific, and there is every indication that this heat has been more or less constant for millions of years in the past. No purely physical theories of its heat are at all satisfactory. The ultimate nature and source of

the sun’s heat are unknown--though various theories have been advanced by way of explanation. I have discussed this question at some length, however in my little book in the present series, “New Discoveries in Science,” to which the reader is referred. (“What Keeps the Sun Hot?”)

The nature and constitution of the sun have, of course, been studied intensively for many years by astronomers. The apparent surface of the sun is called the “photosphere” (light surface). It is the part that gives forth most of the light and heat. Above the photosphere lies a sheet of gas, probably from 500 to 1,000 miles thick, called the “reversing layer,” which is cooler than the photosphere. Outside the photosphere is another layer of gas, from 5,000 to 10,000 miles deep, called the “chromosphere” (cooler sphere). The outermost portion of the Sun is the “corona” (crown). It is a halo of pearly light surrounding the sun, but it cannot be seen except during a total eclipse. It is of irregular form, and gradually fades out into the blackness of space at a distance of from 1,000,000 to 3,000,000 miles. This must not be confounded with the so-called “prominences,” which are vast eruptions of flame, spurting out from the sun’s surface, and extending into space for enormous distances--perhaps half a million miles! These also travel with enormous velocity--five or six hundred miles per second. The earth would appear an insignificant speck of dust, in this vast, roaring furnace of flame!

Of late years, the question of the so-called “sun spots” has aroused a great deal of interest, partly by reason of the fact that they apparently affect electrical and magnetic conditions upon the earth. These “spots” seem to reappear at stated intervals, and about every eleven years reach their maximum intensity, (The period of the revolution of the sun on its own axis has been estimated by their study.)

Although the sun-spots were studied before the eighteenth century, it is only within the past few years that the significance of these enormous spots has become known. For long it was thought that they were merely the great volcanoes of the sun; centers of great heat, generated by the glowing, fiery gases of the sun. In 1908, however, Prof. Hale demonstrated that the sun-spots acted as attraction centers, which drew towards them the hydrogen of the solar atmosphere. “Subsequently, it was found that these spots are the seats of great cyclones, in which cool hydrogen gas is set whirling and is sucked down in the great mælstrom of the Sun, rushing into the center of the spot at the rate of 60 miles a second. Consequently the spots are the center of great solar disturbances, which are of an electromagnetic nature.” From this it was concluded (1) that the spots are cooler than the surrounding area; (2) they are centers of violent cyclones; and (3) they are magnetic fields of great intensity. The connection of sun-spots with our weather, and the relation of one to the other, have also been studied within the past few years.

MERCURY

This is the smallest of the planets in our solar system, being only about 3,000 miles in diameter. It revolves round the sun in almost exactly 88 days, at a distance of approximately 36,000,000 miles, varying between 28½ million to 43½ million miles.

Owing to its smallness, it is often difficult to see this planet, and a powerful telescope must be employed to study it effectually. It is thought that Mercury possesses mountains, but it is practically devoid of atmosphere. Dark, irregular spots have been observed upon the planet, and its surface is thought, by some astronomers, to resemble that of Mars, to a great extent. The solar heat on Mercury is about seven times that on us, owing to its proximity to the Sun. Schiaparelli, and others, have contended that the period of rotation of Mercury is exactly equal to its period of revolution round the Sun. If that be true, one side of the planet is always turned to the Sun, and the other side away from it. One side of the planet would thus be intensely hot, while the other side would be icy cold. Comparatively little is known concerning this small planet, though much study has been devoted to it.

VENUS

This is one of the brightest and most beautiful “stars” in the sky, and has for long been the theme of poets. It is practically the same size as our earth, its diameter being approximately 7,500 miles. It travels round the Sun in 224 days, at a mean distance of 67,000,000 miles. It revolves on its own axis in about 23½ hours, so that its days are nearly the same as ours. It is thought that Venus has a fairly dense atmosphere, and also water-vapor, which proves the existence of water upon its surface. Dense layers of cloud exist in the upper regions of its atmosphere, making direct observations of its surface very difficult. For this reason, comparatively little is known as to the conditions on the surface of Venus. Every eighth year Venus passes through a period of great brilliance, being then so bright as to cast a shadow, like the Moon. Venus must have a higher temperature than our earth; its sky is always overcast; thunder and lightning must be never-ending. Some controversy has existed as to the habitability of Venus; but the consensus of opinion is that life would be practically impossible upon its surface.

THE EARTH

The earth on which we dwell was thought by the ancients to be the center of the entire Universe--the Sun, stars and the vast host of Heaven were thought to revolve around it. This was the so-called “geocentric” theory. It was later on displaced by the so-called “heliocentric” theory, when it was found that the Sun, and not the Earth, was the center of our system. It only remains to be said that our Earth is the third of the planets which revolve round the Sun--Mercury and Venus being nearer the Sun than we are, and the others still further removed in space. We are thus but one of a number of similar bodies moving through space, all revolving round the central Sun.

MARS

This now-famous planet is somewhat smaller than our earth, being about 5,000 miles in diameter. It travels round the Sun in 686 days, at a mean distance of 140,000,000 miles. The eccentricity of its orbit is considerable. Mars appears to us reddish in color, owing to the vast stretches of arid soil (desert) which exist upon its surface.

Water exists, but it is relatively scarce; it is gradually drying-up, as it does in the case of all bodies of the kind--the water on our own earth is very gradually becoming less and less, as the centuries pass.

Mars has been drawn particularly to the public's attention, of late years, by reason of the dispute (still raging) as to its habitability, and the character of its so-called "canals"--the more or less regular markings, like long, dark lines which have been seen to exist upon its surface. Its surface has accordingly been studied minutely for many years, but it may be said that no unanimity of opinion exists as to its being an inhabited planet. Says Kæmpffert, in his "Astronomy," (pp. 183-84):

"The mapping of Mars is no recent matter, for even in 1659 a rough sketch of the surface of the planet was made by Huygens, in which the V-shaped markings at the equator, pointing to the north, can be identified as the _Syrtis Major_. This was followed by rough sketches from time to time down to 1840, when Maedler first began a systematic charting of the planet. His map was followed in 1864 by Kaiser's, by Flammarion's in 1876, and Greene's in 1877. Drawings of various parts of the planet were made during these intervals, but were not combined into good charts."

"Observations made by Prof. Lowell and his staff at the Observatory, at Flagstaff, Arizona, have had a study of this planet especially in view.... The surface of Mars as seen in the telescope, is composed of two white polar caps, which wane with the approach of summer; orange areas, which are supposed by Lowell to be deserts, and blue-green areas, which change their hue to orange during the Martian autumn and winter, and resume their verdant tint in spring. The planet is covered with a network of fine lines, first discovered by Schiaparelli, in 1877; and called by him 'canals'--a designation by which they are still known. These canals connect the polar caps with the temperate and equatorial zones. According to Prof. Lowell, they may be regarded as planetary irrigation ditches, which serve the purpose of leading the melting water of the poles to those desert regions which would still blossom, if properly watered. The canals disappear with the approach of winter, and creep down from the poles towards the equator in summer--a phenomenon which long puzzled astronomers, until Pickering ingeniously suggested that we see, not the canals themselves (for they are much too narrow) but the vegetation which fringes their banks--which withers as the cold of winter descends, and which flourishes with the melting of the snows."

It may be said that this theory of the canals on Mars is not universally accepted by astronomers, but is warmly disputed in some quarters. The _markings_ are undoubted; but some astronomers are inclined to think they are due solely to vegetal growth, and are not the result of human hands. The controversy still continues. Meanwhile, two satellites of Mars were discovered in 1877.

JUPITER

Jupiter is the largest of all the planets, having a diameter of about 88,000 miles; it is only about 1,000 times smaller than the sun--that is, about 1,000 times larger than our earth. In volume, it is 1,300 times larger than our globe. Gravitation must be enormous on its surface. Its density is however relatively low--being only about

one-quarter that of the earth. It is thought to be a world of water and more or less dense gas. It is constantly covered by a thick blanket of clouds and vapors, making direct observation very difficult--as we saw was the case with some other planets. Immense as this planet is in size, it revolves at a tremendous speed--approximately 10 hours. Owing to its immense bulk, it cools more slowly than a body such as our earth. Consequently it will take tens of millions of years for it to cool sufficiently to permit life to become manifest upon its surface. Yet it may at that time! In a sense, Jupiter may be said to be a planet of the future; when our earth is cold and dead, Jupiter may be teeming with animate existence.

Jupiter has 8 satellites, and a number of dark bands cross its surface from east to west. A certain dark spot upon its surface has caused great interest among astronomers, who are unable to determine its exact nature. This planet revolves round the Sun in rather more than 11¾ years, at a mean distance of 483,000,000 miles.

SATURN

Saturn revolves round the Sun in 29½ years, at a mean distance of 886,000,000 miles, in an orbit slightly eccentric. According to Barnard, its equatorial diameter is 76,470 miles, and its polar diameter 69,770, which figures imply a polar compression of 1/11. This planet is famous for the celebrated “ring” which surrounds it. As a matter-of-fact, when observed by means of high-powered telescopes, this famous “ring” is found to consist of a number of rings--three being clearly distinguishable. For long it was thought that these rings were vaporous; then that they were solid; but the present view is that they are composed of myriads of discrete particles of matter, so closely compacted together that to our remote eyes they appear as a solid mass. These rings are fairly broad, but relatively thin in diameter; they resemble a sort of huge disk. Perhaps 100 miles would be the thickness of these rings. Their diameter, however, is tens of thousands of miles in breadth. Owing to these rings, Saturn is one of the most beautiful of all the planets, when viewed through a high-powered telescope.

Saturn doubtless has certain features in common with Jupiter, as to its physical appearance. The general hue of the planet is yellowish-white; it probably has no atmosphere, or at most a very tenuous one. It is attended by ten satellites, the largest of which is known as Titan, thought to be about 2,700 miles in diameter.

URANUS

This is the next to the last planet in our solar system. It is a large planet, having a diameter of about 31,000 miles. Uranus revolves round the Sun in rather more than 84 years, at a mean distance of 1,781,000,000 miles. It is attended by four (or five) satellites. Belts and spots have been seen upon its surface, but relatively little is known concerning its physical conditions, owing to its great distance from us, and its relative smallness. This planet was discovered, as is well known, by Herschel, and was named after him, but its name was subsequently changed. (It is still mentioned

as Herschel, in certain books upon Astrology.) It is probable that the temperature of Uranus is relatively low, owing to the small percentage of the sun's rays which reach its surface. It has been calculated that its theoretical temperature is about 330° F.

NEPTUNE

The past century was remarkable for the discovery of this new planet: Neptune. How it was accomplished is a matter of great interest. In 1820, it was found that Uranus was not following its computed path. Adams, of Cambridge, and Leverrier, of Paris, each independently took up this question, and, assuming that this perturbation was due to the presence of a planet still more remote from the sun (which had been hinted at in 1830 by Bessel) they set to work to calculate its position in the heavens. They finished this at about the same time, arriving at practically the same conclusions. Adams' results were first submitted to the Astronomer Royal, who set them aside without consideration until too late. Leverrier sent his conclusions to a German astronomer, Galle, who found the planet the first evening he looked for it, September 23, 1846.

Neptune is the furthest known planet of our solar system; it is a very large body, having a diameter of about 37,000 miles. It revolves round the sun in an immense orbit which it takes 164 years to complete. Its mean distance from the Sun is nearly 2,800,000,000 miles--a majestic sweep through the heavens! Relatively little is known as to the physical conditions of Neptune, owing to its immense distance, and its relatively small size. At least one satellite is known to exist. The temperature on Neptune must be extremely low, owing to its distance from the sun, whose rays would be exceedingly feeble at that great distance.

THE MINOR PLANETS

Between Mars and Jupiter, a number of small bodies are known to exist, which have sometimes been dignified by the name of "minor planets." They have also been called "Planetoids" and "Asteroids." Special names have been given these individual small bodies--Eros, Ceres, Pallas, Vesta, Juno, etc. Eros passes, upon occasions, very close to our earth; again passing beyond the orbit of Mars. Several hundreds of these smaller planets are now known to exist--as though they were the remnants of some shattered world. These bodies are very small: Ceres, _e.g._, being about 250 miles in diameter; Pallas, 304 miles; Vesta, 211 miles; while a number of the others are thought to be from 5 to 15 miles in diameter.

ARE THERE OTHER PLANETS?

Inasmuch as some of the planets known to us have only been discovered so lately (relatively) the question has naturally been asked: "Why may there not be _other_ planets, beyond Neptune, still undiscovered?" It is a perfectly legitimate question, and no definite answer to this query can be given. It seems rather improbable that another planet will be discovered. However, it is a conceivable possibility, and M. Flammarion has stated that, in his estimation, such a planet probably

exists--gravitating at a distance 48 times as great as the distance between the earth and the sun--that is to say, 7,500 million miles, in an immense orbit which it takes at least 330 years to accomplish. Proof as to the existence of such a planet has not, however, as yet been forthcoming.

THE MOON

The Moon is the Earth’s only satellite, and by far the nearest body in space to our Earth. Many astronomers are inclined to think that the Moon at one time formed a part of the Earth, but was wrenched away from it, leaving a huge cavity, which is now occupied by the Pacific Ocean. The Moon is a cold body, emitting no light or heat of its own; all the light which it seems to shed is entirely _reflected_ light--reflecting the sun’s rays, much as a mirror might reflect them; hence its beautiful silver color.

In round numbers, the Moon is approximately 240,000 miles distant from us in space--the distance varying from 221,600 miles to 252,970 miles--causing a corresponding variation in its apparent diameter and parallax.[A] The circumference of the Moon’s orbit is a little more than a million-and-a-half miles, and it travels through space with the velocity of 2,288.6 miles per hour, or 3,357 feet per second.

[Footnote A: See page 58 for a definition of this term.]

Our satellite always keeps the same face turned towards the Earth, so that we only see one side of it; the other side is forever hidden from the sight of man. However, the axis of the Moon tilts, in relation to the earth, and permits us to glimpse a little more of the surface both north and south, so that about five-eighths of the surface has actually been observed.

The surface of the Moon has been subjected to intensive study, and its “geography” is now as well known as that of our own earth. Vast “seas” (i. e.) sea bottoms, mountain ranges, solitary mountain peaks, enormous craters, are readily observed, and modern telescopes have now brought the moon so close to us that it has been said that any body as high as the Woolworth Building, in New York, would cast a shadow which could be observed and noted.

The fact that the same face of the Moon is always seen by man does not mean that this body remains stationary; it revolves on its own axis, from west to east, but this revolution occupies exactly one siderial month. The result is that the days and nights on the Moon, are many times the length of our days and nights. The surface exposed to the sun’s rays must get extremely hot, and, when deprived of these rays, extremely cold. It has been estimated that the mean temperature of the moon’s surface must approximate 200° F., during the “day” time, and approach the intense cold of inter-stellar space during the “night” (perhaps -250° C.). This would render life or vegetation of any kind very unlikely. However, Professor Pickering has lately asserted that vegetation _does_ apparently spring into being with extreme rapidity during the moon’s day time--evidently remaining latent during the intense cold of the “night.” (Some interesting material on this topic may be found in Shipley’s “Is the Moon a Dead World?” No. 557 of the present series.)

The volume of the Moon is about one-fiftieth that of the earth, but its mass is only about one-eightieth that of our planet. The Moon is practically devoid of atmosphere, which is another reason why it cannot support “life,” in our sense of the word. Two theories have been advanced as to the absence of the moon’s atmosphere: (1) that it gradually combined, chemically, with the materials on its surface; and (2) that it gradually escaped into space, because of the low gravitational pull of the moon. There is no water on our satellite, which means that there is no ice and no snow. The moon being so much smaller than our earth, the pull of gravity is of course much less also.

Yet it is well known that the tides, on the earth, are greatly influenced by the moon. Every atom composing our satellite must exert some subtle pull upon every atom of our oceans, in order thus to affect them. What is the nature of this attraction? Here we encounter the mystery of gravitation! This question must accordingly be postponed until we come to our discussion of that subject.

One of the most remarkable and distinguishing characteristics of the moon consists in the so-called “lunar craters,” which appear to be immense, extinct volcanoes. More than 30,000 of these have now been mapped, varying in size from small hills to immense basins 50, 60, 100 miles in diameter. Ptolemy is 115 miles across, while Theophilus is 64 miles in diameter and 19,000 feet deep. The curious thing about these lunar craters is that they are unlike the hilly volcanoes known to us on our earth. They are rather huge circular pits, often square miles in extent, surrounded by a circular wall, and almost invariably having a single mountainous cone in the center.

Various theories have been advanced by way of explanation of these craters. The most important of these are (1) that they represent extinct volcanoes; (2) that they indicate spots where masses of matter have dashed into the moon, from surrounding space; and (3) that they represent the surface of the moon, when it was a hot, seething mass--their resemblance to the “bubbles” formed at the surface of boiling glue, mud, etc., being pointed to as analogous. Unanimity of view does not exist even yet as to their origin.

The ever-changing “phases” of the moon have been observed by generations of lovers. Thus, the new moon, full moon, etc., are commonplace sights. These apparent changes are, of course, due entirely to the relative position of the sun at the time. If the sun illumines the whole face of the moon, as viewed from our earth, we have full moon; if only a small portion of it, we see the first quarter, etc. The whole disk of the moon may always be seen, however, by careful observation. It is hardly necessary to say that the so-called “Man in the Moon” is a mind’s eye picture, created by the configuration of the various mountains, seas, etc., upon its surface.

THE ORIGIN OF THE SOLAR SYSTEM

Men in every age have speculated as to the constitution and origin of our world, and of the Universe in general. The first really detailed and scientific attempt was made, however, little more than a hundred years ago by Laplace--and subsequently known

as the Laplacian hypothesis (1796).

Concurrent with the establishment of new facts, there was a tendency, throughout the past century, to find some philosophic interpretation of the Universe and its structure; to ascertain, if possible, the “beginnings of things,” and explain them in some satisfactory manner. This has been considered as epoch-making in astronomical research as Darwin’s great theory of the Origin of Species was in biology. The history of the two theories has been similar also. Both have served a useful purpose; have helped to direct scientific thought for years; and both are now largely outgrown. Both were, however, of great value and of daring originality.

Laplace assumed the primal existence of a glowing ball of gas rapidly revolving about an imaginary axis running through its center of gravity. During the process of cooling, this mass would contract, and a disk of gas would be thrown off in this manner; and hence a number of gaseous rings be formed, which would ultimately cool down and assume a spherical form. Laplace conceived that this process might be interfered with by internal accident and by comets from without.

The first modifications of the theory were suggested by Sir Norman Lockyer, who proposed what is known as the meteoritic hypothesis in its place. The central idea of the theory was that--“All self-luminous bodies in celestial space are composed either of swarms of meteorites or of masses of meteoric vapor produced by heat.” The theory was based on spectroscopic analysis. It said that the original nebulæ were composed, not of gases, but of meteoric material and cosmic dust. This theory was never fully accepted in place of that of Laplace, however; but it paved the way for a more recent theory, which may be said to be satisfactory and more or less inclusive. This is known as _the planetesimal hypothesis_, and was advanced within the past few years by F. R. Moulton and T. C. Chamberlin, of the University of Chicago. At the present time, it may be said to be the accepted theory, so far as any such theories are accepted, since it accords with all the facts in a remarkable manner, and has been experimentally demonstrated. In outline, the theory is as follows.

If examination of the nebulæ in the sky be made, out of 120,000 of them, nearly every one of them is found to be in the spiral form. So common and universal is this, indeed, that it was concluded that this must represent “some prevalent process in celestial dynamics.” This process is, according to Chamberlin, the actual formation of a solar system. As this spiral revolves, it accretes to itself various smaller bodies, with their gases, atmospheres, etc., and these become consolidated with the original body. As time went on, this spiral gradually tended to decrease its speed, but at the same time, continued to accrete bodies which came into contact with it in its flight through space. Thus, we have to imagine our world, not as an expanded molten mass which has continuously cooled and contracted, but, on the contrary, as a small lump of cold and solid fragments that, moving about in accordance with its attractions, continuously fed upon its surrounding assemblage of “smaller fry,” and thus grew to its present size. About the young earth so engaged it is possible to read, on the basis of the hypothesis, something of its early history.

Thus we see that the old theory of Laplace has been reversed; and that, instead of a great central mass of moving, white-hot gas, we have a number of smaller bodies, all busily engaged in building up themselves, at the expense of the surrounding masses of still smaller matter--much as a crystal accretes to itself minute specks of crystalline matter from the solution in which it is immersed. This is the newest of the cosmological theories. According to it, all the planets might have been formed at the same time. This view of the formation of the universe opens up still wider problems, which are now the subject of keen debate.

CONSTELLATIONS

The ancients, when studying the heavens, saw all kinds of imaginary animals in the various star-groups, and named them accordingly. A constellation is really a group of stars, which seems to constitute a sort of system of its own. Thus, we find reference to the Great Bear, the Little Bear, the Bull, etc. It is difficult for the uninitiated to see the resemblances which the ancients did, in these various star-groups, and astronomical science has re-named them, as well as adding a large number of new constellations to those already known.

Stars of the first six magnitudes (roughly) are visible to the unaided eye; those of lesser magnitude must be detected by the aid of telescopes. About 5,000 are thus visible; the number is increased according to the magnifying power of the telescope used, and it is estimated that there are more than 100,000,000 within the range of visual and photographic instruments!

The names of a few of the best known constellations are as follows: Ursa Major (The Great Bear); Cassiopeia; Hercules; Scorpio (the Scorpion); Corona Borealis (The Northern Crown); Boötes (The Hunter); Leo (the Lion); Andromeda; Perseus; Auriga (The Charioteer); Taurus (the Bull); Orion; Canis Major (The Great Dog); Canis Minor (The Smaller Dog); Gemini (the Twins), etc.

In these various constellations, certain noted stars are to be found. Thus, in Gemini, its two principal stars are Castor and Pollux. In Canis Major is Sirius. In Orion may be found Aldebaran and Betelgeuse. The Pleiades and Hyades groups are in Taurus. In Perseus is Algol. In Lyra is the first-magnitude star Vega. And so on.

The “Big Dipper,” so-called, is part of the Constellation Ursa Major; and it is almost universally known that the Pole Star (Polaris) may readily be found by its means. The constellations must be traced and learned, one by one; but this the student must accomplish for himself!

METEORS: “SHOOTING STARS”

What are popularly known as “shooting stars” are not stars at all; they are really meteors which appear at altitudes of from 60 to 100 miles, as a rule, from the earth, and move over paths of 40 or 50 miles at a rate of from 10 to 50 miles per second.

The light given out by meteors is due to their being heated by friction with the atmosphere. Falling from space, they become attracted by the earth's gravitation, and fall towards it. Here they encounter the earth's atmosphere, and their rapid passage through it creates terrific heat, which tends to consume them before they reach the face of the earth, turning them into gases, or causing them to fall gently as dust. This sudden flash is the "shooting star" in question.

The number of such meteors is very great. It has been computed that between ten and twenty million strike the earth's atmosphere daily. Occasionally, a large number of meteors fall together; and then we have a "meteoric shower."

METEORITES

Occasionally, however, some of these bodies _do_ reach our earth, despite the friction and opposition of the earth's atmosphere. Such bodies are called meteorites, siderites, or aerolites. Only a few of these are seen to strike the earth yearly, and it is a remarkable fact that, so far as we have any record, not one of them has ever struck a town or killed an individual. The outside of the meteorite during its passage through the air is subject to intense and sudden heating, and the rapid expansion of its surface-layers often breaks it into many fragments. The surface is fused and, on striking, cools rapidly. The result is that it has a black, glossy structure, usually with many small pits where the less refractive material has been melted out. Such meteorites may be seen in most large museums.

COMETS

During the past century, many of these "tramps of the solar system" have been discovered and their orbits computed. The "head" may range from ten thousand to a million miles, or more, while its "tail" may stream across the heavens for millions of miles. These comets' tails always point away from the sun; and for long the reason for this was not known. It is now believed that this is due to light-pressure; the energy of the sun's rays press this delicate matter outwards into space. (This theory has been elaborated at considerable length by the Swedish astronomer and chemist, Arrhenius.)

Many readers of this little book will remember the excitement caused by Halley's comet, which came relatively close to the earth in 1910, so that many persons thought there would be a collision, and were terrified accordingly! As a matter of fact, the tails of comets are usually of almost inconceivable tenuosity. Halley first observed this comet, computed its orbit and predicted the date of its return.

Some comets have tails: others do not. Not much is known concerning the origin and destination of comets; where they originated, or how. They travel at tremendous speed over many millions of miles of space, returning after a few years, or after a lapse of several centuries. They are very striking looking, even when observed by the naked eye. A number of comets have been noted. The following are a few of the more remarkable comets which were observed during the past century:

The Comet of 1811. This was visible for nearly a year and a half, and was carefully studied by William Herschel. Its tail was said to be nearly a hundred million miles long, and fifteen million miles broad.

Encke's Comet (1819). This comet is of extreme interest because of its change of volume. Moulton says: "On October 28, 1828, it was 135,000,000 miles from the sun, and had a diameter of 312,000 miles. On December 24, its distance was 50,000,000 miles and its diameter was 14,000 miles; while at its perihelion passage of December 17, 1838, at a distance of 32,000,000 miles, its diameter was only 3,000 miles."

Beila's Comet (1826). This comet has a most interesting history. In 1846 it was again seen; and a month later it had divided into two parts. They traveled along parallel orbits, some 160,000 miles apart. In 1852, they were seen to be 1,500,000 miles apart. Since then they have never been seen. They have, apparently, vanished from the face of creation!

Donati's Comet (1858). This comet was visible for more than nine months. Its tail was estimated as 54,000,000 miles long. Its period of revolution was more than 2,000 years.

The Great Comets of 1880 and 1882. The latter of these passed through some hundreds of thousands of miles of the sun's corona. Its orbit was not appreciably changed, but, after emerging, it was seen to possess at least five nuclei--showing the effect upon the comet of the disruptive forces through which it had passed.

NEBULAE

These are of especial interest, for the reason that they have played so large a part in forming cosmic theories--the Laplacian, the Planetestimal, etc. Nebulæ are of various kinds--"Annular Nebulæ," resembling a flat, oval, solid ring, having a dark hole in the center. Then there are "Elliptic Nebulæ," of varying degrees of eccentricity; the Great Nebula in Andromeda being a good example. (Numbers of isolated stars may be found within its limits.) "Spiral Nebulæ" are, perhaps, the best known of all, and their name accurately describes their appearance. There are also the so-called "Planetary Nebulæ," as well as Nebulous Stars, Irregular Nebulæ, etc. Of late years, much interest has been centered upon the so-called "Dark Nebulæ." Herschel had long before described various "holes in the heavens," wherein no stars could be discerned. It is now believed that such spots do not represent "holes," as much as dark masses of matter, which seem to blot out the bright stars behind them. The interested reader may refer to Hale's "The Depths of the Universe" for additional information upon this topic, which is relatively new to astronomy. It is also interesting to note that the spectra of Nebulæ contain the bright lines in the green of a substance called "nebulium," because it is not found except in nebulæ.

THE MILKY WAY

This is, in a sense, one vast nebula running right round the heavens in the form of a belt, or ring; its familiar resemblance to spilt milk being the origin of its popular

name. To the naked eye, it appears merely a hazy band of light, but the telescope shows that it is made up of an enormous number of stars, millions of miles apart, but which can only be distinguished from one another by telescopic aid. It constitutes the so-called “Galaxy.” It seems to be spread out in the form of a vast disk, whose diameter is many times its thickness. Our solar system appears to be near the center of this vast system, and, as we penetrate further and further into space, it becomes apparent that fewer and fewer stars, and fewer and fewer nebulæ, seem to exist. Hence the limitation of the material Universe. The Milky Way is made up of thousands of millions of suns; yet their enormous distances make them appear to constitute one vast, luminous belt encircling our globe!

THE NUMBER OF STARS

When the heavens are viewed with the naked eye, a few hundred stars may perhaps be seen--some bright, some faint. Viewed through opera glasses, many more stars may been seen; while their number is again greatly increased by the use of a telescope. The larger and more powerful the telescope employed, the greater the number of stars thus discovered in the depths of space. The interesting question thus arises: What is the total number of stars in the entire firmament? Can they be estimated? And if so, what would their approximate number be?

What we call “stars” are, of course, in practically all cases _suns_--often vastly larger and hotter than our own. These stars differ from one another in order of brilliance; some are brighter than others. They are accordingly classified according to their order of brilliance, and known as stars of the “First Magnitude,” of the “Second Magnitude,” etc., up to about the Seventeenth Magnitude. Any star of a given magnitude is, roughly, about two-and-a-half times as brilliant as one of the next lower order, and this variation holds throughout--each magnitude being that much greater in brilliance.

The “magnitude” of the stars varies according to their light-giving power, and also their distance from us. One of the methods adapted to measure the magnitude is to compare its brightness with an artificial star, gradually cutting-off its light by means of neutral, tinted glass until the two are equal. The color of the star must be taken into account, in such measurements, the eye being more sensitive to some colors than to others.

Now, it is an interesting and significant fact that the number of the stars decreases as their magnitude decreases; that is to say, the greatest number of stars are found of the first magnitude; a lesser number of the second; still less of the third, and so on (broadly speaking). After reaching the ninth magnitude, the number very rapidly diminishes. It has been calculated that there are about 120,000,000 stars in the first 16 or 17 magnitudes. If the proportion were maintained throughout, however, there would be more than ten times that number. Some authorities have asserted that there are, roughly, half a billion stars of varying magnitudes in the heavens.

THE POSITION OF OUR SOLAR SYSTEM

It has been maintained that our solar system is at, or very near, the center of the whole Universe. Certain it is that the further we proceed into space, the less the number of stars encountered, which has given rise to the suspicion that their number is actually _limited_, and that the whole Universe consists of a sort of sphere, in which is enclosed all the stars that exist, and that, beyond this sphere, no stars whatever remain. No matter exists beyond this point! Such a view fits in rather well with Einstein's conception of "curved space," and a finite universe of infinite proportions! Of course, it is conceivable that, outside this vast system, another similar system may exist, and another and still another; but of such systems we know nothing, and it seems improbable that proof of their existence could ever be obtained by man. So far as we can tell, the universe is _One_, and the matter and energy of that one are limited.

THE MOVEMENT OF OUR SOLAR SYSTEM

It has been shown that our whole solar system is sweeping through space at the speed of about ten miles a second towards the stars in the constellation Hercules, and particularly towards Vega, one of its suns. However, Vega is likewise moving through space, so that by the time our sun reaches the spot now occupied by Vega (half a million years or so) Vega will no longer occupy that position, and no "collision" will take place in consequence! We shall not, in fact, pass very near that star.

DISTANCES OF THE STARS

Astronomical distances are so vast that they can only be measured in the mind relatively. The distances between the planets in our own solar system seem big enough; yet they shrink into insignificance when compared to the distances which separate our whole solar system from even the nearest of the stars. _Alpha Centauri_ is the nearest star, and it is separated from us by a distance 276,000 times as great as that which separates us from our sun. It is approximately 25 billion miles away. Traveling with the speed of an express train flung into space, at 40 miles an hour, towards the nearest star, without any stoppage or any slowing down, we should not arrive at our destination until after an interrupted flight of 75 million years. Yet this is our nearest neighbour! Only a _very few_ of the stars are within 400,000,000,000,000 miles of the sun. The great majority of them are many times this distance from us.

So vast are these distances that some simple means of expressing them on paper was sought. A "light Year" was finally decided upon as the _unit_ of measurement--that is, the distance which light would travel in one year, speeding at the rate of 186,000 miles a second. It has been estimated that many stars are one, two, three and perhaps five hundred thousand light-years distant from us in space. The interested reader may figure-out the number of miles this represents for himself!

TEMPERATURE OF THE STARS

Measurements which have been undertaken prove that the surface temperature of our Sun is between 5,000°C. and 7,000°C. It is thought that many stars are considerably hotter than this. We can form no adequate conception of such intense heat; all matter would be vaporized; yet, under the enormous pressures which must prevail, these vapors would in turn be converted into thick, semi-fluid substances--especially in the interior.

FIXED STARS

The so-called “fixed” stars are those which do not appear to change their positions in the heavens for long periods of time together. There are, of course, no “fixed” stars at all since every celestial body is moving with greater or lesser rapidity through space; but these stars are so far distant from us that such movements are inappreciable, even after long periods of time, and in spite of the most careful observations. In comparison with the more rapidly moving heavenly bodies, they do not appear to “move,” and have been denominated “fixed stars” in consequence.

DOUBLE STARS

A large number of stars appear single, when viewed by the naked eye, but when seen through a powerful telescope, are seen to be, in reality, two stars which revolve round one another. Many thousands of such double stars are now known to exist; indeed, apparently single stars have been found, upon closer examination, to be composed of a group of four or five or more stars--so that the name “multiple stars” has been given to such groups. They are near one another in the astronomical use of that word--though they may actually be hundreds of thousands, or millions of miles apart. Many of these double stars seem to be quite separate from one another. Others appear to have some physical connection. Those which are known to form systems are known as _binaries_.

COLORED STARS

Many of the double stars exhibit curious and beautiful phenomena of complementary colors. In such cases, the larger star is usually more or less reddish or orange, and the smaller one bluish-green or greenish-blue. Many of the double stars, on the contrary, are of the same color. There are white, red, blue, orange, green and yellow stars. The planets also vary greatly in color--Venus, e.g., being white, Mars reddish, etc. Inasmuch as the planets only reflect light, however, this is due to quite different causes; the other colored stars are self-luminous suns which emit light of their own.

VARIABLE STARS

In addition to variations in the color of stars, they also vary greatly in brilliance, and certain stars are much brighter at times than at others. In some cases these changes in brilliance are regular; in others, irregular. “Omicron,” for example, which, Bayer recorded in his Atlas in 1603, is a regular variable; its period of change is 331 days, 8

hours; in other words, it reaches its greatest brightness about 12 times in 11 years, when it sometimes attains the brilliancy of a star of the 2nd magnitude, at which brilliancy it remains stationary for about a fortnight. It then diminishes during about three months, until it sinks down to a star of magnitude 9½, or even becomes totally invisible. It remains in this condition for about 5 months, and then gradually recovers--during the next following 3 months--its maximum brilliancy. In other words, its brilliancy is absolutely periodic. Other variables are by no means regular, however, but “come and go” at different intervals.

Various theories have been advanced by way of explanation--one of the simplest being that such stars are in reality double, one being luminous and the other not; and that, during their revolutions, the non-luminous star partially or totally eclipses the bright one, at stated intervals. The whole subject, however, is difficult, and much yet remains to be learned concerning these variable stars.

TEMPORARY STARS

From time to time, stars have suddenly appeared in the heavens, where no star existed before! Such stars have usually become increasingly brilliant for a short period of time, and then as suddenly died away again, leaving no trace of their existence behind them. These “new stars” for long puzzled astronomers. The theory often advanced to explain them is that some distant star has “exploded,” and the increasing brilliance which we see is the result. If such were the case, its sudden dimming-down and disappearance would be quite intelligible--as would be its sudden appearance. A large number of such stars have now been recorded, and their existence is no longer in doubt. In some cases, they have remained visible for weeks or months before their final disappearance.

STAR GROUPS--CLUSTERS

Here and there throughout the sky are places where the brighter stars seem to be clustered. These families of stars are of such magnificent proportions as to stagger the imagination. Among the best known are the Pleiades, the Hyades, Coma Berenices and Orion. Although they appear to us very close together, they are not really so, being usually several hundreds of thousands of miles apart. Many of these star-groups are irregular; but numbers of them constitute clusters, which are of various sizes and shapes. Perhaps the most interesting are the so-called “globular clusters,” because they present the appearance of stars having been massed together as globes. Some of them contain five or six thousand stars. Although they appear to us so close together, it has been calculated that, in a cluster containing 5,000 stars the average distance of the stars from one another would be 30,000 times the distance of the sun from the earth! The vast distances of space considered in astronomy may perhaps be realized by this fact--when it is considered that such a cluster appears to us as a single star, only capable of being separated into its component parts by means of high-powered telescopes!

ECLIPSES

The total eclipse of the Sun, January 24, 1925, brought the subject of eclipses to the public attention as never before, and many thousands of persons watched that beautiful and impressive sight through smoked glasses or strips of film.

When we speak of eclipses, we usually mean an eclipse of either the Sun or the Moon. How are such eclipses caused?

A total or partial eclipse of the sun is caused by the moon passing between the earth and the sun, the three celestial bodies forming, as it were, a straight line. The sun is then shut-off from the vision of the inhabitants of our globe over a certain, limited area of its surface. The shadow cast by the moon falls across the earth.

But how is the moon eclipsed? Certainly the sun does not pass between the moon and the earth, on such occasions! What causes the moon to be eclipsed?

The answer is as follows: Inasmuch as both the earth and the moon are illuminated by the sun, they both cast long shadows into space, as any solid body does, when held in front of a strong light. The earth’s shadow trails away for thousands of miles into space. Into this shadow the moon enters, and when it does so, it becomes eclipsed--totally or partially, as the case may be. Total eclipses are instances when the whole surface of the celestial body is apparently covered; partial eclipses are those in which only a portion of the body is dark--the remainder being still visible.

In addition to eclipses, two other astronomical phenomena of interest should here be mentioned: _Transits_, and _Occultations_. By “transit” is meant the passage of some other heavenly body between ourselves and the sun. Thus, Mercury and Venus, both lying nearer the sun than the earth, occasionally pass in front of it. We then have a transit of Venus, or a transit of Mercury, as the case may be.

By “Occultation” is meant the hiding of one heavenly body by another--as when the moon hides some other planet or star, or one planet hides another planet or star. The three bodies are then “in line” as before. Of course, all eclipses represent instances of Occultation.

TELESCOPES

Telescopes are of relatively recent origin; the ancients were forced to make their observations without them, which makes some of their conclusions all the more remarkable. There is considerable evidence that the builders of the Great Pyramid employed the “Grand Gallery” for astronomical observations (see “The Great Pyramid of Egypt,” in the present series), and other devices were employed. But no telescopes of any great power of magnification existed before the last century, while our present marvelous instruments of precision are the evolution of the present century.

Telescopes are of two kinds: refracting and reflecting. Any small telescope exemplifies the former; the incoming light-rays are focussed by a series of lenses, and directly observed by the eye. In the employment of reflecting telescopes, however, another principle is employed: the incoming light-rays are caught and reflected by

means of a curved mirror, and focussed on a lens, which in turn is inserted in an elaborate eye-piece, in which the light-rays are magnified and measured. Some of the modern instruments have a forty or more inch aperture, and are capable of enormous powers of magnification.

THE SPECTROSCOPE: SPECTRUM ANALYSIS

For more than two thousand years, astronomy remained a purely mechanical and mathematical science, being limited to observations and deductions therefrom; but in 1860 the method of spectrum-analysis was discovered. This was a most revolutionary discovery, inaugurating, as it did, the whole science of astro-physics; and enabling us to know as much of the physics and chemistry of distant stars and nebulæ--their nature, constitution, and temperature--as we know of the planets of our own system! Even the existence of otherwise invisible stars has been demonstrated in this manner--their orbits, rate of motion, and mass. The science of astro-physics is now one of the most exact in the whole realm of science; and has only been rendered possible by the invention of the spectroscope. As this instrument plays such an important part in all astronomical research, a brief explanation of the instrument becomes necessary.

If a ray of sunlight be passed through a glass prism, the ray is split up into its primary colors; so that, instead of a single spot of white light being visible a narrow band of brilliant colors is seen--ranging from red to violet. But this is not the most important part of the discovery. When this spectrum was closely examined, it was found to be crossed by numerous black bands of various thicknesses. Sometimes these occurred in groups, sometimes singly. By enlarging the spectrum by passing it through several prisms, as many as 3,000 of these bands could be counted. The nature and explanation of these strange bands of blackness remained long uninterpreted, however. It remained for Kirchoff, in 1860, to discover their uses and significance.

Briefly, it is this. The chemical elements, when heated to a state of incandescence, present each one its own characteristic spectrum; each one has its own peculiar markings, or band of lines. No two elements are exactly like in their bands, as shown in the spectrum. Hence, whenever that particular marking is observed, it becomes certain that that element, and none other, is present. These spectra are very varied; iron, for example, has more than 2,000 such bands, while lead and potassium have but one each.

In this way--all the chemical elements having been studied, and their characteristic bands known--it became possible to explore the stars, planets and suns, and discover their chemical composition. For, no matter where an element was discovered--on this earth or on the remotest star--it would always cast its particular spectrum, when thus examined. The effect of all this upon astronomy can be perceived at once. Not only the heavenly bodies known to us, but those which have never been seen by human eye--even when aided by the most powerful telescopes--can be studied and their chemical composition and structure accurately determined. Here is progress indeed!

All this becomes the more remarkable when we stop to consider the immense distances of space, and how widely separated the heavenly bodies are from one another. This may, perhaps, be shown by one or two illustrations. We are, roughly, about 93,000,000 miles from our own sun. Now, the majority of the stars we see are suns, like ours. The sun next removed from us in space is about 275,000 times as far from us as we are from our sun. The orbit of Halley's comet, of which so much has been written lately, is some 3,280,000,000 miles in length; and this sporadic body, coursing through space at a speed 50 times greater than a rifle bullet, takes 75 years to complete its circuit. The nearest star has been calculated to be nearly 25 trillion miles away; while some of the stars are 40 times as far from us as that!

PHOTOGRAPHY

The second great engine of astronomical research, that has been added during the past century, is _photography_. By this means exact maps may be taken of the heavens at any hour of the night, and the precise position of thousands of stars determined with the utmost exactitude. A chart of the heavens, made in this manner, is not only more complete but more accurate than the combined observations of any number of men could possibly be. Moreover, the photographic plate will record the existence of stars which cannot be seen even with the aid of the most powerful telescopes. This is due to the fact that the plate gradually collects light, and its _cumulative_ effect is noticeable, when its _immediate_ effect cannot be perceived. This power of photographic plates is most valuable, and cannot be duplicated in any other manner. We are assured on good authority that "an ordinary good portrait camera with a lens three or four inches in diameter, if properly mounted so that an exposure of several hours can be made, will show stars so minute that they are invisible even in the great Lick telescope." An international photographic chart of the heavens is now under way, which, when finished, will represent an accurate catalog of every visible sun, star, and planet, in the sky. After this, any unusual body should be quickly discovered.

But photography is employed not only for mapping out the heavens, but for reaching the farthest stars. The moon and the sun have both been photographed repeatedly, and with most instructive results. The first good pictures of the moon were made by Dr. John W. Draper of New York City, in March, 1840. His son, Dr. Henry Draper, succeeded him in this work, and his photographs were considered the best until Rutherfurd began his remarkable work in 1865. After this, much important work was done in the Lick observatory, and elsewhere. The first picture of the sun was taken in 1845, by Fizeau and Foucault, on a daguerreotype plate. Sun spots, total eclipses, etc., are now studied in great detail by this means.

THE TIDES

Every particle of matter attracts every other particle of matter throughout the entire Universe. The Sun and the Moon both exert a definite pull upon the earth; the moon particularly, being the earth's satellite, is (so to say) held in place by the earth. The

moon, exerting this definite pull, naturally influences the water of the earth most of all, because water is a fluid, mobile body. A heaping-up of the water then occurs--“high tide.” But the moon also attracts the earth to some extent; and the consequence of this is that the water on the opposite side of the globe is, as it were, left behind, which causes a heaping-up of the water there also. Hence, there are two high tides daily, with an interval of 12 hours between them, on opposite sides of the globe.

When the sun and moon pull together, we have the highest tides--“spring tides.” When they do not pull together (being in different parts of the heavens) we have only the surplus pull of the moon over the sun, and the tides are consequently not so high. These are the “neap tides.” All tides act as a sort of check or brake upon the rotation of the earth on its axis--tending to slow down its speed to some extent. “Tidal waves” are due to a combination of special causes.

GRAVITATION

The mysterious influence or “pull” which various celestial bodies exert upon one another is known as gravity or gravitation. We know that masses of matter attract one another according to their size; the larger the body, the greater the force exerted, etc. Further, the influence decreases according to a definite law--according to the square of the distance between the two bodies. The innermost nature of gravitation is still largely a mystery--though various ingenious theories have been advanced in order to explain it. (See my article in “The Monist,” for July, 1913, and pp. 44-46 of “New Discoveries in Science” in the present series.) Gravitation is supposed to act throughout the whole Universe, so that all celestial bodies mutually influence one another, to some extent. Its speed, mode or action, etc., as well as its essence or true nature are, however, unknown even yet; they are still unsolved mysteries!

THE ETHER

At all events, gravitation is thought to act through, or by means of, the Ether--the nature of which is still another mystery! Lodge, in his “Ether of Space,” has given some interesting figures as to the enormous strain which the ether must be supposed to transmit or carry. Lack of space, however, prevents a further discussion of this interesting question; a brief summary may be found on pp. 53-55 of my book on “Chemistry for Beginners,” in the series of Blue Books. For our present purposes, it need only be said that the ether is the only hypothetical connecting-link between celestial bodies--since there is no air or atmosphere in interstellar space. And it is across or by means of this ether that gravitation must be exerted.

ATOMIC ANALOGIES

Recent investigations of the innermost structure of the atom have shown us that it is probably constituted on very much the same plan as our solar system--a central “sun”

or proton, round which revolve the negative planets or “electrons.” This question I have treated more fully in my “Chemistry for Beginners,” pp. 42-44, to which the reader is referred.

THUNDER AND LIGHTNING

The lightning flash is merely a huge electric spark, such as may be seen between the terminals of any electric machine. In cases of flashes, or forked lightning, this “spark” is seen directly. Sheet lightning is observed when the original flash is hidden behind clouds, and only its reflection or effects are seen. The rumbling of thunder is due to the reverberations and echoes of the original “peal.” The peal is thought to be due to the sudden rushing together of the molecules of the upper atmosphere, which have been rent asunder by the flash--a sort of vacuum created. Camille Flammarion has written an interesting book on “Thunder and Lightning,” which may be consulted for further details.

FIREBALLS

These are virtually the same as “shooting stars” (_q.v._,) and no essential difference can be pointed to, as to their origin or nature. They are not mere “blobs” of lightning, but solid bodies which sometimes burst, with a great noise--though they are usually noiseless. Many of them appear to be pear-shaped, but they may be seen to change their size and shape during the period of visibility. Fireballs are often accompanied by a train of sparks.

ATMOSPHERIC ELECTRICITY

The surface of the earth is constantly charged with negative electricity of a static character. The upper atmosphere is usually charged positively, though, this may vary according to circumstances. The earth and upper air thus resemble two sheets of tin-foil, with the air an imperfect dialectric between them. This may be broken down, especially in wet or damp weather. The effects upon the mental and physical health are often very noticeable (see Dexter: “Weather Influences,” etc.)

THE EARTH’S MAGNETISM

It has long been known that the magnetic pole does not coincide with the North Pole (or South Pole). The compass points to the magnetic north pole, and not to the true north pole. Lines of magnetic force seem to envelop the earth, terminating at the north and south poles, respectively. Although this is purely a terrestrial phenomenon, it is necessary to mention it here, since it has enabled us to explain, very largely, the remarkable manifestation known as

THE AURORA BOREALIS

This is usually seen in northern climes, and the reason for this is now clear. We know that the corpuscles discharged from a Crookes tube are deflected by a magnet. These corpuscles are discharged in immense numbers by the sun, and rain upon our earth. Now, the earth is a magnet, and these corpuscles are caught by the lines of force girdling our earth, and carried towards the poles, where they find themselves in an atmosphere comparable with high vacua. They then begin to give out the shifting and darting lights characteristic of the cathode rays, causing a certain luminosity. These darting and shifting lights would, on this theory, account for the Aurora Borealis--which is also known to vary with the number of sun-spots.

TIME: MEASUREMENT OF:

Our divisions of time are purely arbitrary, and are all based upon the revolution of our earth upon its axis, which thus constitutes a gigantic clock. All other clocks, watches, etc., are adjusted accordingly. This is really our only way of measuring time; subjective feelings are very illusory, and have to be checked-up by other means. The solar day is the basis of all our calculations--a month, a year, etc., being only so many days in length. Our earth, therefore, is the clock by which we measure the time of the Universe!

SPACE: MEASUREMENT OF:

The measurement of space is always a difficult problem, even for near-by objects (see my “Psychology for Beginners”). When applied to celestial bodies, it becomes immensely complicated, and the only wonder is that such apparently accurate measurements have in fact been made! Such measurements cannot, of course, ever be made _directly_, but must depend upon trigonometry and abstruse mathematical calculations. Most of them are based upon the following principles: If we observe a distant object from two different points-of-view, at a known distance apart, the angle formed by imaginary lines running from the object to one position, and to the other, can readily be calculated. Knowing this angle, much can be ascertained as to the size, distance, etc., of the distant body. If a distant star be viewed from opposite sides of the earth, we have here a known base-line of slightly more than 8,000 miles. But this is altogether too small for astronomical distances! A much longer base-line must be sought. Accordingly, observations are made of a distant star when the earth is (so to say) “north” of the sun, and further observations of the same star when the earth is (so to say) “south” of it--six months later, when the earth has traveled half-way through its orbit round the sun. The diameter of the earth’s orbit being known (186,000,000 miles, almost) we have here a base-line of this size for use in our measurement of the angle and subsequent calculations. Immense as this base-line is, however, it is too small for our purposes, for so immense are astronomical distances, that _no change whatever_ can be observed in the relative positions of certain fixed stars--even when studied from such different positions in space! In other words, the star is so far distant that, when viewed from two positions in space, distant from one another nearly one hundred and eighty-six million miles, it appears to occupy the same position! But a mere summary of this question, and its details would involve an

entire volume in itself!

THE INTERNATIONAL DAY LINE

Inasmuch as our earth revolves on its axis, a new day is beginning at some different moment all round the world. This being the case, how are we to fix some definite and official “starting point” for our day--since the day officially begins at midnight, and not at sunrise? To determine this, an arbitrary International Day Line has been drawn, on the 180th meridian--just half way round the globe from Greenwich. Fortunately, this falls in the Pacific Ocean, where there is almost no land. When the sun crosses this line, a new day begins. I have explained this more fully in my book “New Discoveries in Science” in the present series (pp. 40-42).

CALENDARS, ETC.

Our year is a little more than 365 days in length--in fact, nearly 365¼. Because of this fact, an extra day accumulates every four years; and to include this we add this extra day to February every “leap year.” In this way, our celestial bookkeeping is kept fairly accurate. Twelve months of 30 days each would give 360 days, with five days over. It was, however, found that five days was not enough, while five and a quarter was too much. It is interesting to note that Hipparchus, who flourished in the 2nd century B. C., worked on this problem, and fixed 5 days and 55 m., as the time required--a truly remarkable achievement, since it has since been found to be accurate to within less than six minutes.

CURVED SPACE

This, and various other problems connected with the Einstein theories may be found treated in No. 408 of the present series, “An Introduction to Einstein,” by William F. Hudgings.

THE TEMPERATURE OF SPACE

The Earth is warmed by the sun’s rays, some of which are absorbed, while some are reflected. But these rays themselves possess no “heat”; they are merely minute vibrations in the ether. Heat is only present when they strike some solid body. Consequently the vast inter-stellar spaces are tremendously cold--probably at or about absolute zero (-273.10°C). Our earth is not heated directly, as a man is heated by standing in front of a blazing fire; but only by means of electro-magnetic undulations, which traverse millions of miles of space, colder than death, without heating them!

LIGHT IN SPACE

Space is also intensely dark; no light exists there save the faint twinklings of distant stars. The sun illumines our earth, because its rays are reflected from its surface; but space itself is intensely black, just as it is intensely cold It is a “cold world” indeed,

once we have stepped off the little planet on which we dwell!

LIFE IN SPACE

All this being so, life in any form cannot very well exist in space--since the conditions for its existence are altogether absent. Arrhenius has, however, suggested, that the “germs of life” might possibly be carried across millions of miles of space on dust particles, propelled by the energy of light. This, however, is a pure theory, which has so far received no official proof.

THE CAUSES OF AN ICE AGE

We know that our Earth has passed through several ice ages, in the past, and various astronomical theories have been advanced in order to explain this fact. Perhaps the most ingenious of these is that advanced by Sir Robert Ball (see his “The Cause of An Ice Age”). Very briefly, it is that the eccentricity of the earth’s orbit and the tilting of the polar axis causes an ice age, or the reverse. If the northern axis is tilted towards the sun, when nearest to it (so to say), then the northern hemisphere will enjoy a genial climate, and if the southern axis be thus tilted, the reverse conditions will prevail. This, and various other theories have, however, been discussed by Finger in his book on “The Ice Age,” in the present series, No. 327.

WHY DO STARS “TWINKLE”?

When we look at a star near the horizon, we at once notice that it twinkles, or “scintillates,” especially in the winter time. The phenomenon is purely atmospheric, and is due to waves of air of unequal density sweeping across the line of sight. When viewed through a telescope, this is sometimes magnified into actual dancing.

WHY DOES THE MOON SOMETIMES APPEAR LARGER?

It is well known that the moon often appears larger when rising or setting--i. e., near the horizon, than when it is overhead. The same is true of the Sun. It is hardly necessary to say that these celestial bodies have not _actually_ increased or decreased in size! Why, then, should we perceive them larger at some times than at others?

The reason for this is two-fold; psychological and optical. In the first place, the Heavens do not appear to us quite round, but somewhat flattened out, like a watch-glass. Hence the moon appears to be much further away when it rises than it does when it is overhead, with nothing between. The moon near the horizon is apparently larger because it seems further away. The second reason is that the refraction of the earth’s atmosphere gives this illusion of increased size.

ARE THE PLANETS INHABITED?

This is a much-disputed point! Various astronomers (Schiaparelli, Lowell, etc.) have contended that they have almost indubitable evidence that Mars is inhabited by living beings like ourselves; other equally competent astronomers assert the contrary. Certainly, none of the planets of our own solar system, with the possible

exceptions of Mars and Venus, could possibly be inhabited. That is universally granted. And we have no _direct_ evidence of any other inhabited worlds throughout space. Analogy, however, forces us to believe that, of the millions of suns blazing in the heavens, many of them must be attended by a planetary system such as ours; and if such be the case, there is no reason why life should not originate and thrive thereon as well as upon our own planet. We have, however, no means of proving or disproving this directly.

In our own system, Venus and particularly Mars offer possibilities. Venus probably always turns one face towards the sun, so that this side would be tremendously hot, while the other side would be frozen in perpetual ice. Mars is a possibility; and, as we know, great controversy has raged regarding the habitability of this planet, and as to its “Canals.” The interested reader may refer to Lowell’s “Mars as the Abode of Life,” and “Mars and Its Canals” for the affirmative, and to Maunder’s “Are the Planets Inhabited?” for the negative, side of this question.

A FEW DEFINITIONS

What “Parallax” means. Since the earth revolves round the sun, the stars are apparently in slightly different directions from it at different times of the year. The difference in direction of a star as seen from two points on the earth’s orbit which are separated by the mean distance to the sun is the _parallax_ of the star. In other words, the parallax of a star is the angle subtended by the major semi-axis of the earth’s orbit, as seen from the star.

The “Orbit” of a moving body is its more or less circular passage through space, usually around another larger body, as our earth revolves round the sun. The “eccentricity” of the orbit consists in the fluctuations or variations from its exact path.

The “Ecliptic” System. If we could see the stars near the sun, we should find that the Sun apparently moves eastward among them, completing one revolution in a year. Tracing such a path, it will be found that it more or less coincides with the celestial equator. The equator and the ecliptic intersect at two points; these points are the “equinoxes” the _vernal_ equinox being the one at which the sun crosses the equator from south to north, and the _autumnal_ equinox the other one.

“Satellites.” These are smaller bodies which revolve round large ones, and, so to say, attend them. All except two of the planets are known to have satellites revolving round them, just as they revolve round the sun. Mercury and Venus have none; the earth has the moon; Mars has two little moons, only a few miles in diameter; Jupiter has four large satellites and four small ones; Saturn has ten, one of which is larger than Mercury; Uranus has four satellites, and Neptune one.

The “Planetoids.” Between Mars and Jupiter a number of small bodies have been discovered, moving in a regular orbit; these have been called planetoids. If some planet has once occupied this mid-way position, and subsequently exploded, the fragments would occupy the position occupied by the planetoids. Whether or not this

is their origin is a disputed point, which it would take us too far afield to consider here. They suggest the possibility.

“Planets.” These are the bodies revolving round a central sun. Aside from those constituting our own solar system, we see no planets in space; we see suns, or stars; but if the latter have planets attendant upon them, we cannot see them.

The point of the moon’s orbit nearest the earth is called the _perigee_; the furthest point, the _apogee_.

TRANSCRIBER’S NOTES

Author’s spelling of “dialectric” has been retained.

Inconsistencies in hyphenation have been left unchanged.

Typos corrected:

Title page: missing opening quote in “Psychology for Beginners”

Page 13: “concenses” to “consensus”

Page 17: “equitorial” to “equatorial”, "Myriads" to "myriads" (lowercase)

Page 49: “on my book” to “of my book”

Page 54: “litle” to “little”

Page 59: “Unanus” to “Uranus”